NAGDEV PRODUCTS PRIVATE LIMITED
Manufacturer of Castor Oil & Specialty Oleochemical Derivatives · Gujarat, India
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TECHNICAL DATA SHEET
Doc Ref: NPPL-TDS-23
Date: October 2026
Status: Advance Specification

Lithium 12-Hydroxystearate (Lithium 12-HSA)

Proprietary Grade: NDEVSOAP· LI12 | CAS No: 7620-77-1 | Category: Under Development

Physical Form:	Fine Powder / Flakes	Raw Material:	12-Hydroxystearic Acid (12-HSA),
Typical Colour:	White to Off-White	Shelf Life:	24 months in sealed, moisture-pr
Standard Packing:	20 / 25 kg HDPE-Lined Paper / PP Bags · 500 kg HDPE Jumbo Bags · 1,0		

GUARANTEED ANALYTICAL SPECIFICATIONS

SR	TEST PARAMETER	SPECIFICATION LIMIT	STANDARD METHOD
1	Appearance	White to Off-White Fine Powder / Fla	Visual
2	Lithium Content, %	2.0 · 3.0	ICP-OES / AAS
3	12-Hydroxystearic Acid Content, %	85.0 Min	GC / GLC
4	Precursor Fatty Acid Value, mg KOH/g	175 · 190	AOCS Cd 3d-63
5	Hydroxyl Value, mg KOH/g	145 · 165	AOCS Cd 13-60
6	Melting Point, °C	210 Min (Decomposition)	Standard Laboratory
7	Moisture, %	1.0 Max	ASTM E203
8	Free Fatty Acids, %	2.0 Max	AOCS Ca 5a-40
9	Ash, %	10.0 Max	ASTM D482
10	Water-Insoluble Matter, %	1.0 Max	Standard Laboratory
11	Free Alkali, %	0.5 Max	Standard Laboratory
12	Particle Size, Passing Through 100 Mesh, %	95 Min	Sieve Test
13	Penetration of Grease Prepared from Produc	As per Grade / Customer Requirement	ASTM D217

PRODUCT APPLICATION & CHARACTERISTICS

Lithium 12-Hydroxystearate is primarily used as a thickener / lubricating soap for lithium-based greases. It provides good mechanical stability, water resistance and high-temperature performance and is used in automotive, industrial, bearing and general-purpose lubricating grease formulations.

IMPORTANT NOTICE & COMMERCIAL AVAILABILITY:
Commercial production begins in 2027 at our new integrated manufacturing facility in Vav-Tharad, Gujarat.
Advance bulk inquiries, long-term supply agreements, and 500g/1kg R&D lab samples are currently accepted.